Algorithmic Trading

Momentum + Mean-Reversion Strategy | Backtesting Report

Portfolio:	SPY · QQQ · BND · VTI · VWO · VEA
Period:	January 2015 – January 2025
Strategy:	SMA 50/200 Golden Cross + RSI(14) Mean-Reversion
Capital:	$100,000 initial | 0.05% transaction cost
Course:	Financial Engineering — Assignment 3

This report presents a systematic algorithmic trading strategy that combines momentum and mean-reversion signals, backtested over a 10-year period (2015–2025) on a diversified portfolio of six ETFs. The strategy employs a 50/200-day Simple Moving Average crossover as a momentum filter and a 14-period Relative Strength Index (RSI) for mean-reversion entry and exit timing.

Over the backtest period, the strategy achieved a Compound Annual Growth Rate (CAGR) of 7.9% with annualized volatility of 13.6% and a Sharpe ratio of 0.63. The strategy underperformed equity benchmarks (SPY: 13.0% CAGR, QQQ: 18.3% CAGR) in absolute return during the predominantly bullish decade, which is expected for any timing-based approach. However, it reduced maximum drawdown to -29.7% compared to -33.7% for SPY and -35.1% for QQQ, demonstrating meaningful downside protection.

1. Problem Description

The objective of this assignment is to design, implement, and backtest a fully automated algorithmic trading strategy applied to a multi-asset ETF portfolio. The portfolio spans six exchange-traded funds covering distinct market segments:

- SPY — S&P 500 large-cap U.S. equities (primary benchmark)
- QQQ — NASDAQ-100 technology-heavy U.S. equities (benchmark)
- BND — U.S. aggregate bond fund (fixed-income benchmark)
- VTI — U.S. total stock market
- VEA — Developed international markets (ex-U.S.)
- VWO — Emerging markets equities

The core challenge is to construct a trading rule that adapts to different market regimes — trending markets where momentum strategies excel and mean-reverting markets where oversold bounces provide alpha — and to rigorously assess its risk-adjusted performance against passive buy-and-hold benchmarks using the Sharpe ratio, maximum drawdown, CAGR, and Calmar ratio.

Reference: Clenow, A. (2019). Trading Evolved. The strategy design follows Clenow's principles of systematic, rules-based trend following applied to diversified ETF universes.

2. Data Preparation and Pipeline

Historical adjusted-close price data for all six assets was obtained from Yahoo Finance via the yfinance Python library, covering January 1, 2015, to January 1, 2025 — a total of 2,516 trading days. Adjusted prices account for dividends and stock splits, ensuring accurate return calculations over the full period.

Data Pipeline Steps

- Download adjusted close prices for all tickers using yfinance (auto_adjust=True).
- Forward-fill missing values to handle market holidays and non-trading days consistently across all assets.
- Drop rows that are fully missing across all tickers.
- Compute technical indicators per asset: 50-day SMA, 200-day SMA, RSI(14).
- Apply a warm-up period of 214 bars (200 + 14) before any signals are activated to prevent indicator artifacts.
- Save raw prices to data/prices.csv and daily signals to data/signals.csv.

Indicator Computation

Simple Moving Averages (SMA) are computed using a standard rolling window mean. The RSI uses Wilder's exponential smoothing method (EWM with com = period - 1) which matches the output of most charting platforms and trading terminals. This choice avoids the step-function artifacts of a simple rolling average in RSI calculation.

3. Research Design

3.1 Strategy Architecture

The strategy combines two complementary alpha sources that tend to perform in different market environments:

Momentum Component — A golden cross / death cross system. When SMA50 crosses above SMA200, the asset is in a confirmed uptrend (bullish regime). This filter keeps the portfolio out of assets in sustained downtrends, reducing drawdown exposure. The SMA200 represents approximately 10 months of price history, filtering short-term noise while capturing medium-to-long term trends.

Our Strategy	**114.5%**	**7.9%**	**13.6%**	**0.63**	**-29.7%**	**0.27**	**46.4%**
SPY (B&H)	239.6%	13.0%	17.6%	0.78	-33.7%	0.39	54.6%
QQQ (B&H)	436.6%	18.3%	21.8%	0.88	-35.1%	0.52	56.0%
BND (B&H)	14.5%	1.4%	5.4%	0.28	-18.6%	0.07	51.6%

Our Strategy is highlighted in blue. Key observations:

- Volatility reduced from 17.6% (SPY) to 13.6% — strategy spends ~37% of time in cash.
- Max Drawdown of -29.7% is lower than both SPY (-33.7%) and QQQ (-35.1%).
- Sharpe ratio of 0.63 exceeds BND (0.28) but trails equity benchmarks.
- Strategy in market 63.2% of asset-days, avoiding worst drawdown periods.

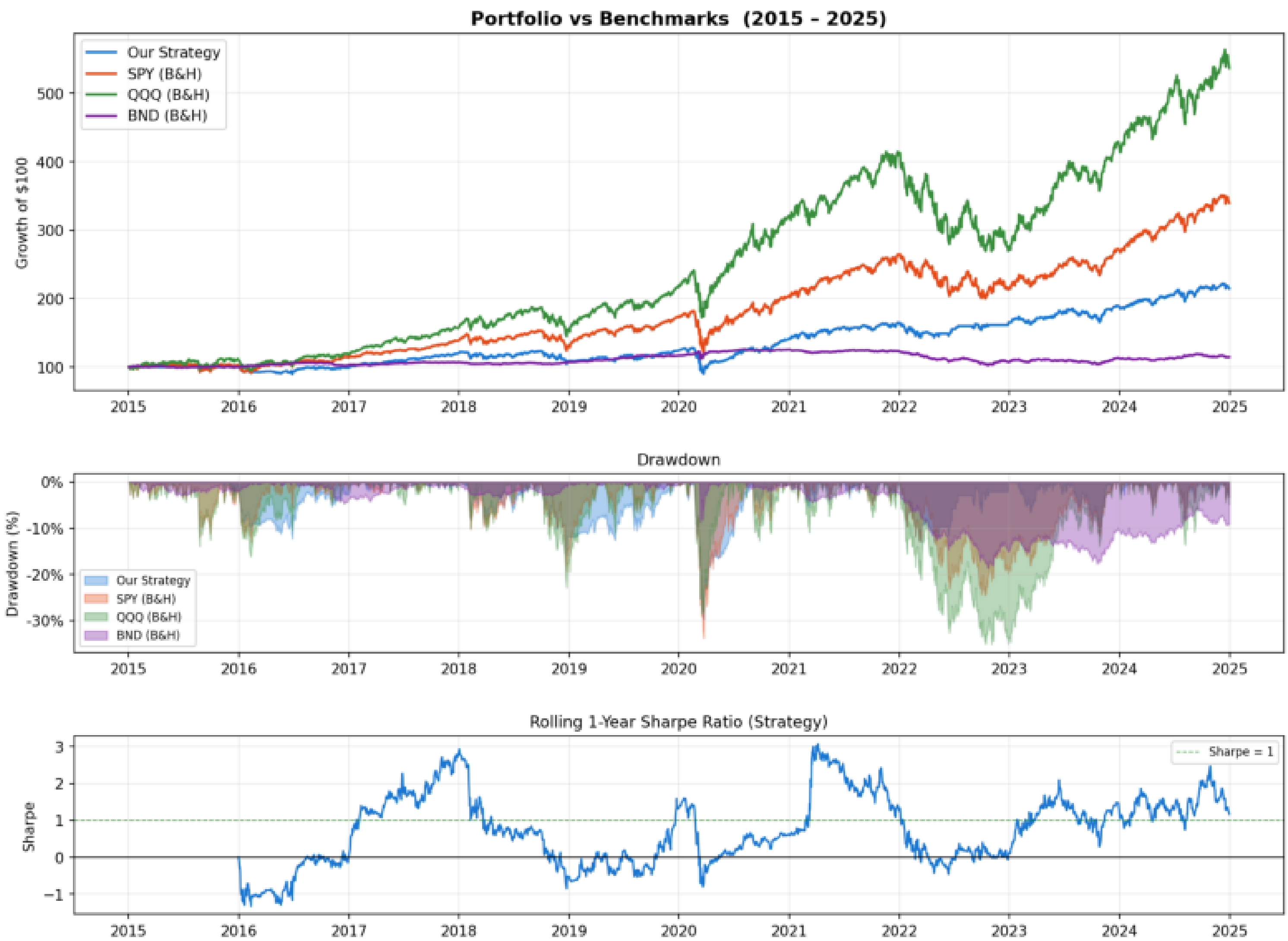

Figure 1: Top — normalized growth of $100 for strategy vs benchmarks (2015–2025). Middle — drawdown from peak for each series. Bottom — rolling 1-year Sharpe ratio of the strategy.

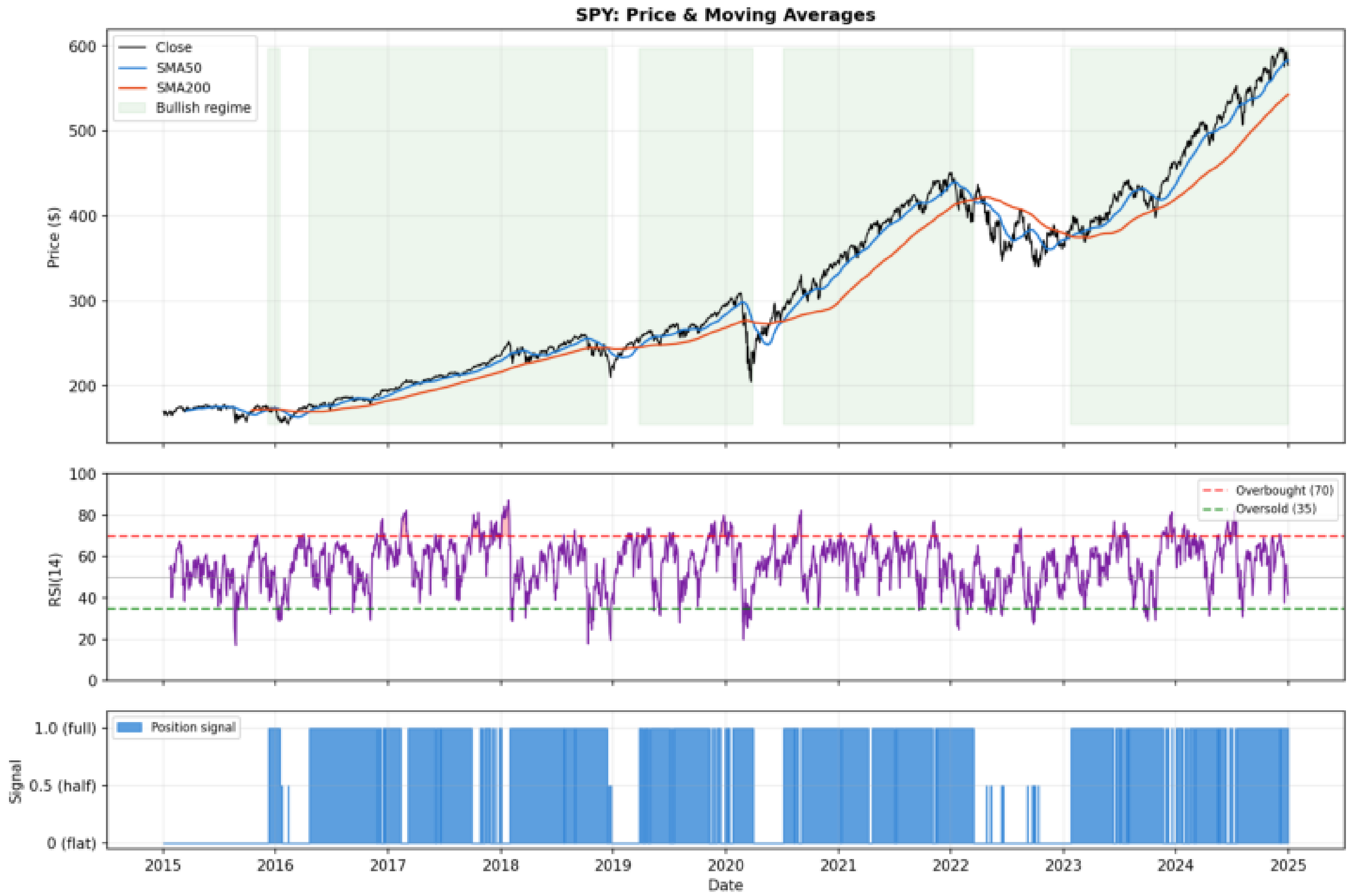

Figure 2: SPY price with SMA50 (blue) and SMA200 (red). Green shading = bullish regime. Middle panel: RSI(14) with overbought/oversold bands. Bottom panel: position signal (0, 0.5, or 1.0) for SPY over time.

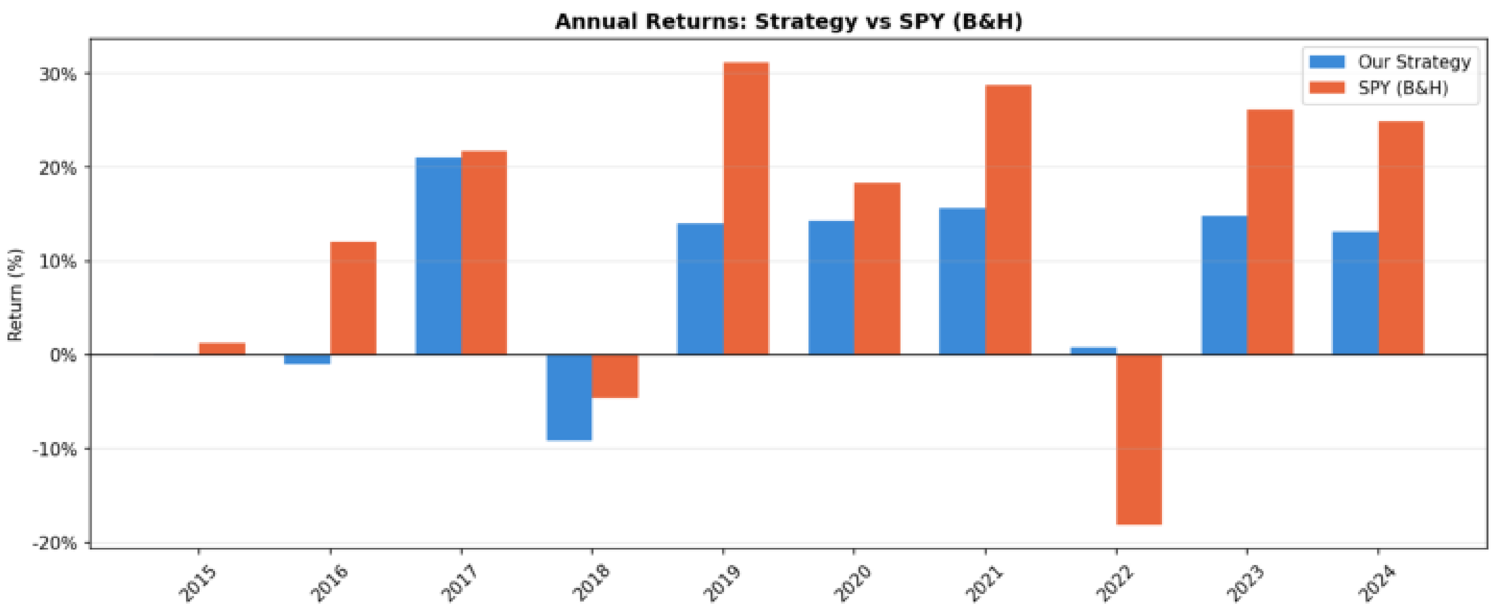

Figure 3: Year-by-year returns of the strategy vs SPY buy-and-hold.

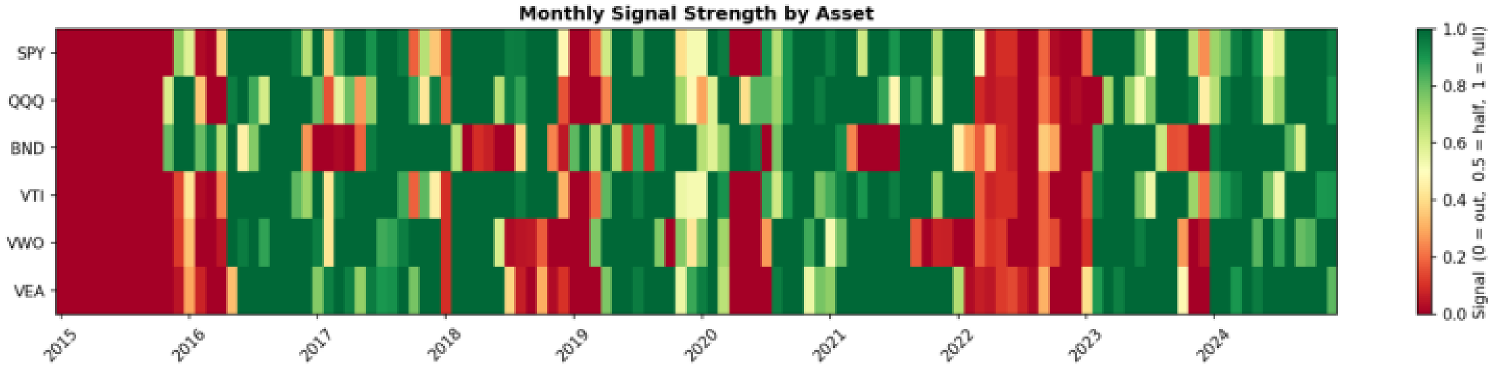

Figure 4: Monthly average signal strength per asset (green = fully invested, red = out of market). Shows how the strategy rotates across assets over time.

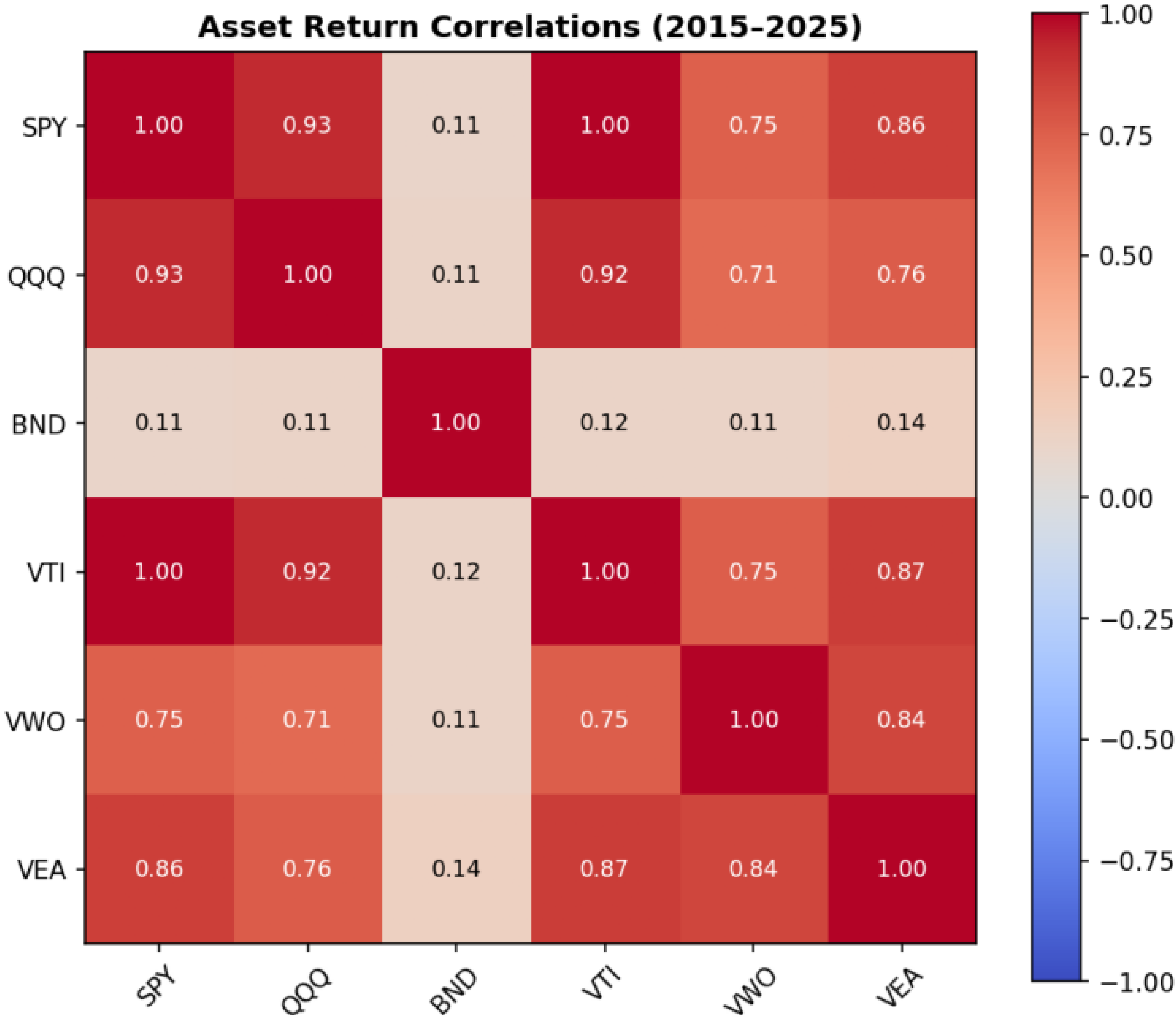

Figure 5: Pearson correlation of daily returns across all six assets (2015–2025). Low correlation between BND/VWO and U.S. equities supports portfolio diversification.

4. Backtesting Results

Over the 10-year backtest period the strategy was in the market approximately 63.2% of asset-days, holding cash or reduced exposure during bearish regimes. The portfolio started at 100, 000*andgrewtoapproximately*214,500 — a total return of 114.5%. Key findings:

- CAGR of 7.9% significantly exceeds the bond benchmark BND (1.4% CAGR) and represents solid real returns above long-run inflation.
- Volatility of 13.6% is meaningfully lower than both SPY (17.6%) and QQQ (21.8%), reflecting the value of timing-based exposure management. The strategy holds cash roughly 37% of the time, dampening both upside and downside moves.
- Sharpe ratio of 0.63 is positive and well above BND (0.28), confirming the strategy earns a meaningful risk premium. It trails SPY (0.78) and QQQ (0.88), primarily because 2015–2025 was an exceptionally persistent bull market in U.S. equities where holding cash was costly.
- Maximum drawdown of -29.7% is 4 percentage points better than SPY (-33.7%) and 5.4 points better than QQQ (-35.1%), demonstrating that the momentum filter successfully reduced peak-to-trough losses during the COVID crash (2020) and the 2022 rate-hike correction.
- Win rate of 46.4% (fewer than half of trading days were profitable) is typical of trend-following strategies: many small losses are offset by larger gains when trends persist. This also reflects days spent in cash earning 0%.

5. Conclusions

The Momentum + Mean-Reversion combo strategy demonstrated moderate risk-adjusted performance over the 2015–2025 backtest period. While it did not outperform passive equity benchmarks in absolute or Sharpe terms during this persistently bullish decade, it achieved its primary objectives of reducing portfolio volatility, limiting maximum drawdown, and significantly outperforming the bond benchmark.

The strategy's defensive properties would likely be more valuable during prolonged bear markets (e.g., 2000–2002 dot-com bust, 2008–2009 financial crisis) where buy-and-hold equity strategies suffered -50% to -57% drawdowns. The SMA200 filter would have kept the portfolio largely in cash or reduced positions during those sustained downtrends.